
STAGE 1
Input Tensor
Raw Trajectory
x in R^24
Process + Patch
STAGE 2
Train Scaling
Fit Scaler on
Normal-Only Train
Train statistics only
STAGE 3
Core Model
Bottleneck Autoencoder
24 -> 16 -> 6 -> 16 -> 24
LayerNorm + ReLU
STAGE 4
Scoring s(x)
Standardized Residuals
Mean Squared Error
Dimensionless Score
STAGE 5
Threshold tau
Val-Tuned Cutoff
Cost-Sensitive F2
validation F2 cutoff
STAGE 6
Operational Alert
Binary Decision
y_hat = 1[s(x) >= tau]
Human Trace Review